THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM THREE ANNUAL EXAMINATION
033 BIOLOGY

Time: 3 Hours **November 2025**

Instructions

1. This paper consists of sections **A**, **B**, and **C** with a total of eleven questions.
2. Answer all questions in sections **A** and **B** and two questions from section **C**.
3. Section A carries sixteen marks, section B carries fifty-four marks, and section C carries thirty marks.
4. All drawings must be in pencil.
5. All communication devices and any unauthorized materials are not allowed in the examination room.
6. Write your Examination Number on every page of your answer sheets.

SECTION A (16 MARKS)

Answer all questions in this section.

1. For each of the items (i)–(x), choose the correct answer from the given alternatives and write its letter beside the item number in your answer booklet.

   (i) A student at Seypalm secondary school was examining a certain plant and was able to observe the following features: Presence of two cotyledons in the seeds, tap root system, and vascular bundled arranged in a ring form in the stem. Which of the following is likely to be the observed plant?

   A. Cactus plant
   B. Grass
   C. Sisal plant
   D. Maize plant
   E. Coconut plant

   (ii) The following are plant excretory products, except:

   A. Tannins
   B. Urea
   C. Oxygen
   D. Carbon dioxide
   E. Latex

   (iii) The main function of the placenta in a pregnant woman is to:

   A. Produce hormones that prevent miscarriage.

B. Protect the foetus from mechanical injury.
C. Facilitate the exchange of materials between mother and foetus.
D. Store nutrients for the growing foetus.
E. To protect ovaries from injury

(iv) What is the process that takes place in humans when subjected to a very cold weather?
A. Vasodilation
B. Sweating
C. Lowering in the metabolic rate
D. Vasoconstriction
E. Flattening of body hairs

(v) A person who has blood group AB can donate blood to a person with blood group:
A. A only
B. B only
C. AB only
D. O only
E. AB and O

(vi) The main site of reabsorption of water in the nephron is the:
A. Glomerulus
B. Bowman's capsule
C. Proximal convoluted tubule
D. Loop of Henle
E. Ureter

(vii) The hormone responsible for the conversion of excess glucose into glycogen in the liver is:
A. Adrenaline
B. Insulin
C. Glucagon
D. Thyroxine
E. Testosterone

(viii) The type of immunity that a baby receives from its mother through the placenta is:
A. Acquired immunity
B. Artificial immunity
C. Natural active immunity
D. Natural passive immunity
E. Vaccination

(ix) In a food chain, the decomposers get their energy from:
A. Producers
B. Consumers
C. Both producers and consumers
D. The sun
E. Plants

(x) The type of joint found at the elbow is a:
   A. Ball and socket joint
   B. Hinge joint
   C. Pivot joint
   D. Gliding joint
   E. Immovable joint

2. Match the items in **List A** with their corresponding descriptions in **List B**.

LIST A	LIST B
(i) Urethra (ii) Alveoli (iii)Homeostasis (iv)Transpiration (v) Pancreas (vi)Small intestine	A. The site where digestion and absorption of food occur B. The final channel through which urine is excreted from the body. C. The process of maintaining a constant internal environment. D. The process of losing water in the form of vapour from plants. E. Tiny air sacs in the lungs where gaseous exchange occurs F. A gland that produces both digestive enzymes and hormones. G. The process of releasing metabolic waste products

SECTION B (54 MARKS)

Answer all questions in this section.

3. (a) Your friend suffered from an electric shock while you were in the school laboratory. Outline four first aid measures you would deliver to her.
   (b) Give five reasons why proper waste disposal is important for human health and the environment.

4. (a) What is malnutrition?
   (b) A woman visited a food seller and bought the following food items: eggs, Irish potatoes, and sunflower seeds.
      (i) Identify the main nutrient found in each food item.
      (ii) Give four functions of the nutrient found in eggs.

5. (a) Explain four differences between breathing and aerobic respiration.
   (b) Outline five factors that affect the rate of gaseous exchange in humans.

6. (a) Briefly explain three methods of disease prevention and control.
   (b) Differentiate between antiseptic and disinfectant, giving one example for each.
   (c) State three examples of zoonotic diseases.

7. (a) With the aid of a well-labelled diagram, describe the reflex arc.
   (b) Explain the role of the two hormones produced by the pancreas.

8. A constant internal temperature permits efficient functioning of the body. Classes of organisms have various control mechanisms for ensuring their internal temperature remains constant. Taking mammals as an example, explain five mechanism they use in controlling overheating and four mechanisms they use in controlling overcooling.

SECTION C (30 MARKS)

Answer any two questions from this section.

9. Justify the statement that "Angiosperm is an important division of the kingdom plantae " by giving and explaining six advantages and two disadvantages of the division.

10. (a) Explain three ways in which a plant is adapted for transporting water from the roots to the leaves.
    (b) Elaborate on three features that make the human heart an efficient pump for blood circulation.

11. (a) Describe five events that occur in a typical menstrual cycle in humans.
    (b) Explain five factors contributing to an increased rate of sexually transmitted infections (STIs) among young people.